

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element L Initial Template AT - 6 points

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

In K3, you provided the general lessons you learned related to engineering design. Now, you are asked to provide recommendations for continuing work on this project.

Answer these questions with **specific recommendations** based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design-specific decisions you made).

Provide recommendations to someone doing a similar project that might build upon or continue your work. For example: "Taking into account <something we did>, we believe doing _____ would improve _____ because _____."

- What could be done next to go further with this project?

Something that could be done next to go further with this project could be getting some expert feedback to help us finalize our design. This could improve the prototype and address weaknesses. This could also help us see where something went wrong during any of the design processes. Something else that could be done is testing the prototype under certain conditions. This could help us see how safe it is.

- What would you keep if you did the project again? **Generate a consistently detailed list** of what would be kept in the prototype or testing plans. (4-5 sentences.)

Something that we would keep if we were to do this project again would be the idea of the robot having heat at the bottom of it. We say this because this would have melted the ice way faster instead of dispensing salt, which we would take way longer. Something that we would specifically keep is the heating cable, metal, and heating cable. This would

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help us solve the issue, but we would need to do more research on how much heat we need so that we wouldn't mess up the robot, but also have enough heat to melt the ice.

- What would you change in your current prototype or testing plan? **Generate a consistently detailed list of HOW these details would be implemented. (4-5 sentences.)**

Something that we could change in our testing plan is testing using different temperatures and humidity levels. We would do this to see how the ice forms and what temperature we would need for the ice to melt. Also, something we could change in our testing plan is measuring the ice thickness. This could have helped us to see how slippery the ice is and would help evaluate de-icing methods. Lastly, we could've done during the testing plan to see how smooth the concrete is, so that we could see how it affects the surface.

- What would have made this project easier for you to complete? **Generate a consistently detailed list of things that would make the project easier. (You may make a list or use 4-5 sentences.)**

Having enough research and materials would have made this project easier for us to complete. We needed this to build the second prototype and make corrections. Research would have made this easier to do because we would have known the different components that we needed to create the prototype. Taking the time to research would have made this process way easier instead of creating something, then having to go back and research and repeat that multiple times. Having the materials would have been easier because we would just have to build using the sketches that were created instead of having to use technology to build it, especially since most of us didn't know how to use it.



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Step L: Presentation of Designer's Recommendations						
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
L1. Content of recommendations. Recommendations include _____.	Specific and detailed ways the project can be improved, and a rationale for why they are needed	Specific and detailed ways the project can be improved	Some specific ways the project can be improved	one way the project can be improved	vague recommendation(s) for improvement of the project	no recommendations for improvement of the project
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
L2. Detailed plans for implementation of improvements. Plans for implementation of improvements are _____.	consistently detailed	generally detailed	minimally detailed and may include one or more caveats	partially detailed and/or overly general	vague/unclear or minimally related to the recommendations provided	not offered